Lesson 3-5 · Period 2

Multiplicity and Factored-Form Graphing

Klimsara-adapted · Student-centered · No Blooket

Do Now — Find the zeros from factored form

Algebra 2 · Lesson 3-5 · Period 2

DOK 1 | 5 min | *bridge from Monday*

Practice (3-5-savvas-q13) · Sketch from zeros Sketch the graph of the function by finding the zeros:

$$g(x) = (x+3)(x-1)(x-4)$$

For each factor: what zero does it give? Mark each zero on your sketch. Use the sentence frame: "The zeros are $x =$ ____, $x =$ ____, and $x =$ ____ because each factor equals zero."

Launch — Multiplicity Rules (Example 2)

DOK 2 | 12 min | *teacher models*

Multiplicity Rules — keep on your page all period

1. If a factor appears **ODD** number of times → the graph **CROSSES** the x-axis at that zero.
2. If a factor appears **EVEN** number of times → the graph is **TANGENT** (touches and turns back) at that zero.

Example (Savvas Ex 2): $f(x) = (x-2)^3(x+1)^2$

- $x = 2$: factor appears 3 times — ODD → **crosses**
- $x = -1$: factor appears 2 times — EVEN → **tangent**

Count the factor. Odd or even? That's your answer.

Explore — TPS (Think-Pair-Share) I

DOK 2 | 33 min | *student work*

5 items in order. For each zero: write multiplicity + crosses/tangent before sketching. ~6–7 min each.

Try It 2a · Describe zeros $f(x) = x(x+4)(x-1)^4$ — describe graph behavior at each zero.

Try It 2b · Real zeros only $f(x) = (x^2+9)(x-1)^5(x+2)^2$ — which zeros are real? Crosses or tangent?

Explore — TPS (Think-Pair-Share) II

Practice #14 · Factor first $f(x) = x^3 - 8x^2 + 16x$ — find zeros, then describe graph behavior.

Practice #15 · Factor by grouping $g(x) = x^3 - x^2 - 25x + 25$ — find zeros, then describe graph behavior.

Practice #16 · Biquadratic $f(x) = 9x^4 - 40x^2 + 16$ — let $u = x^2$; find all zeros.

Share / Summary

DOK 2 | 5 min | *academic conversation + bridge prompt*

Sentence frame · Pick one of your zeros "For $f(x) =$ ____________, the zero $x =$ ______ has multiplicity _____,
so the graph ______________ the x-axis at that zero."

Bridge to Wednesday (Period 3 Do Now) A cubic has zeros 2, $1+\sqrt{3}$, and $1-\sqrt{3}$.
What is the nature of its coefficients — rational, irrational, or complex?
Write the key term "conjugate pair" in your packet margin.
(Directly prepares Topic 3 Assessment Q#2.)

Exit Ticket — Summary Recap

Algebra 2 · Lesson 3-5 · Period 2

DOK 1-2 | 2 min | *not graded*

Complete on your exit ticket

1. Today I learned that ______________________________.
2. The rule I want to remember is __________________________.
3. One thing I'm still unsure about is _______________________.